

MathGov v12.6 / SGP v8.5 and MHIOS v0.7

Capability Claim Integrity and Publication-Readiness Audit

11 August 2026

Core: MathGov_Core_2026_09_v12.6_SGP_v8.5+2026.08.11.1

MHIOS: MHIOS_v0_7+2026.08.11.1

Same semantic versions | Correction build only | Artifact verification PASS

Executive verdict

The requested improvements were implemented without changing the MathGov cascade, formulas, thresholds, rights protections, Union Scopes, Welfare Dimensions, SGP floors, or component semantic versions. The substantive clarification is a MathGov-native Capability Claim Integrity discipline inside Reality Grounding and MFDI. Publication defects were corrected across the primary DOCX/PDF surfaces, especially the Canon and SGP contents pages.

FINAL VERIFICATION: PASS. Construct validity remains UNTESTED.

Capability Claim Integrity implementation

A material capability claim must now be operationalized before it may acquire gate, authority, deployment, conformance, or public-communication weight.

Required field	Function
Operational definition	Measurable functional meaning for the pinned configuration and domain.
Observable criteria	Observations that support or fail the claim.
Mechanism or system composition	Components and interactions instantiating the capability.
Evidence surface	Traces, trials, records, or measurements supporting the claim.
Validity conditions	Configurations, domains, contexts, and controls under which the claim holds.
Alternative explanations	Scripting, intervention, memorization, benchmark fit, proxies, or other material alternatives.
Falsification or revision trigger	Evidence that forces narrowing, reopening, or refusal.
Evidence maturity stage	Boundary between definition, computation, implementation, empirical contact, qualification, and

Required field	Function
	outcomes.
Permitted claim	Strongest wording supported by the evidence.
Explicit nonclaims	Unsupported inferences that are expressly prohibited.

The rule blocks both anthropomorphic overclaim and composition reductionism. A configured system may possess a bounded operational capability through coordinated inference, memory, planning, feedback, tools, deterministic constraints, and human controls. That capability does not establish consciousness, moral agency, generality, safety, legality, legitimacy, or authority.

Exact substantive changes

Surface	Improvement	Status
Canon v12.6	Capability Claim Integrity, operational-label rule, decomposition rule, and Appendix AI clarification.	Implemented; version unchanged.
MFDI v2.3	Capability profile, bounded nonclaim, required fields, and claim-strength examples.	Implemented; version unchanged.
Agent System v12.4	Requires the split record and blocks unsupported stronger claims.	Implemented; version unchanged.
ripple.md v5.5	Carries the decomposition into assurance and public-claim records.	Implemented; version unchanged.
Cascade v2.6	Preserves capability, warrant, objective, authority, and control as separate positions.	Implemented; version unchanged.
Primer v4.4	Reader guidance against scalar autonomy ladders and composition reductionism.	Implemented; version unchanged.
SGP v8.5	Bounded capability neither proves nor disproves sentience or moral status.	Implemented; version unchanged.
Run-record v4	Schema fields, positive vector, and adversarial vectors.	Build contract updated only.
MHIOS v0.7	Object record, Rule 8.0G, anti-theater pattern, schema validation, and coverage.	Implemented; version unchanged.

Publication-format audit

TOC disposition

Document	Final disposition	Entries	PDF location
Canon v12.6	Complete static linked TOC with dot leaders and physical page numbers.	70	Pages 2-3
SGP v8.5	Complete static linked TOC after the reader-facing output map.	28	Page 3
WDBIP v1.6	Complete static linked TOC; front matter repaired.	27	Page 2
MHIOS v0.7	Complete static linked TOC for standard and appendices.	33	Page 2
Other primary documents	Incomplete automatic TOC shells and content controls removed.	0	Not applicable

The Canon and SGP TOCs are now useful, complete, and aligned to rendered PDF pages. No document retains an empty or misleading Word TOC control.

Front matter and placement

- WDBIP no longer shows a YAML-like block, visible page-break marker, or duplicate title.
- The Agent System self-audit comparison boundary is in its substantive section rather than the contents area.
- ripple.md appendix blocks are in their correct appendices.
- All primary DOCXs use versioned footers with dynamic page fields.
- MHIOS date is synchronized to 11 August 2026 across Markdown, DOCX, PDF, VERSION.yaml, and release evidence.

Tables, graphics, and legibility

Metric	Core	MHIOS	Combined
Primary documents	15	1	16
PDF pages	596	68	664
Tables	287	17	304
Table rows	2,439	198	2,637
Headings	1,634	246	1,880

Metric	Core	MHIOS	Combined
Linked TOC entries	125	33	158
Comments	0	0	0
Tracked changes	0	0	0
Content controls	0	0	0
Clipped/out-of-page text	0	0	0
Blank PDF pages	0	0	0
Tiny-font spans below 6 pt	0	0	0
Drawings/charts/image blocks	0	0	0

All 304 tables have explicit borders and repeating first-row header properties. Multirow tables use the restrained light-blue header hierarchy. Six one-row WDBIP record tables intentionally remain unshaded because they do not contain a separate header/body matrix.

No decorative graphic was added. This is deliberate: the governing specifications are definition-, rule-, record-, and table-intensive. A new diagram would not add semantic precision and would create an additional accessibility and synchronization surface.

Accessibility and PDF preflight

Audit	Core	MHIOS	Result
High-severity accessibility findings	0	0	PASS
Medium-severity accessibility findings	0	0	PASS
Low-severity findings	76	0	Bounded
PDF preflight warnings	0	0	PASS

The 76 Core low-severity findings are raw URL display text in scholarly reference lists. They were retained because explicit DOI and institutional URLs are useful citation information. There are no heading-order, table-header, missing-alt-text, or broken-navigation findings. MHIOS has zero accessibility findings.

Verification results

MathGov Core

Surface	Result
Master verifier	PASS
Subordinate verification surfaces	9/9 PASS
Core run-record vectors	6 positive / 29 expected failures
SGP RMCP vectors	1 positive / 1 expected failure
WDBIP vectors	1 positive / 17 expected failures
Aligners Sheet	87 worksheets / 1,643 formulas / 0 errors
Core 15 mirrors	Exact byte-identity PASS
Full render and visual QA	596/596 pages reviewed

MHIOS

Surface	Result
Package verifier	PASS
Objects / roles / screens	46 / 15 / 20
Minimal Vertical Slice screens	5
Conformance vectors	4 positive / 52 expected failures
Seeded mutation suite	40/40 killed
Atomic modal clauses	158
Any executable linkage	67 / 158 (42.41%)
Direct behavioral linkage	57 / 158 (36.08%)
Explicitly unlinked clauses	91 / 158
Main standard	68 pages / 17 tables / 33 TOC entries

The increase from 154 to 158 modal clauses is expected: Capability Claim Integrity adds four genuine normative requirements. The increase from 65 to 67 any-executable and 55 to 57 direct-behavioral links reflects added schema and validator coverage. The 91 unlinked clauses remain visible bounded open work.

Version and build disposition

Artifact	Semantic version	Final build identity
MathGov Core	v12.6	MathGov_Core_2026_09_v12.6_SGP_v8.5+2026.08.11.1
SGP	v8.5	Included in the Core build
MHIOS	v0.7	MHIOS_v0_7+2026.08.11.1

No component semantic version was incremented. The new build identity is required because package bytes, navigation, schemas, tests, reports, and hashes changed.

Explicit nonchanges

- No sixth gate.
- No change to RG -> RF/NCRC -> TRC -> CSV -> RLS.
- No change to seven Union Scopes or seven Welfare Dimensions.
- No change to numerical defaults, rights non-compensation, TRC/CSV separation, or SGP protection floors.
- No imported external ontology, formula, taxonomy, verdict system, hardware architecture, implementation logic, or proprietary term.
- No claim that determinism establishes truth or that probabilistic methods are intrinsically invalid.
- No claim of empirical superiority, legal approval, physical-safety certification, consciousness detection, construct validity, or deployment readiness.

Final readiness disposition

MathGov Core v12.6 / SGP v8.5 is complete and publication-ready as an open research-source governance specification. MHIOS v0.7 is complete and publication-ready as a candidate implementation companion for controlled development and pilot research. The documents are legible, navigable, internally consistent, reproducibly identified, and fully verified at the artifact and tested-interface levels.

The next legitimate work is empirical: independent implementation, domain calibration, inter-rater reliability, adversarial testing, burden and usability measurement, institutional validation, and outcome-based requalification.